



Science — Cbse Half-Yearly Exam — 80 Marks

ANSWER KEY — FOR GRADING

Subject: Science | Class: Class 9

Total Marks: 80 | Time Allowed: 3h | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 20 questions. Each carries 1 mark(s).		
1	Which of the following correctly pairs a connective tissue with its connection? (A) Tendon - connects bone to bone (B) Ligament - connects muscle to bone (C) Cartilage - connects bone to muscle (D) Tendon - connects muscle to bone Answer: Correct Option: (D) ► Why? D. Tendons connect muscles to bones, while ligaments connect bones to bones. [Ch 3: Tissues in Action]	1
2	If a plant is unable to transport food from its leaves to its roots, which specific tissue is most likely malfunctioning? (A) Xylem (B) Epidermis (C) Phloem (D) Sclerenchyma Answer: Correct Option: (C) ► Why? C. Phloem is the complex permanent tissue that transports food to all parts of the plant, so its malfunction would hamper food transport. [Ch 3: Tissues in Action]	1



Q.NO.	QUESTIONS	MARKS
3	In which year did Isaac Newton present his three laws of motion? (A) 1607 (B) 1687 (C) 1707 (D) 1757 Answer: Correct Option: (B) ► Why? B. Isaac Newton presented his three laws of motion in the year 1687. [Ch 6: How Forces Affect Motion]	1
4	Which of the following is classified as a water-in-oil emulsion? (A) Milk (B) Butter (C) Vanishing cream (D) Sugar solution Answer: Correct Option: (B) ► Why? B. Butter is a water-in-oil emulsion, where water is the dispersed phase and oil is the dispersion medium. [Ch 5: Exploring Mixtures and their Separation]	1
5	A sound wave of frequency 20 kHz propagates in air with speed 344 m s^{-1}. What is its wavelength? (A) 17.2 m (B) 17.2 cm (C) 1.72 cm (D) 0.172 m Answer: Correct Option: (C) ► Why? C. $\lambda = v/\nu = (344 \text{ m s}^{-1})/(20000 \text{ s}^{-1}) = 0.0172 \text{ m} = 1.72 \text{ cm}$. [Ch 10: Sound Waves: Characteristics and Applications]	1
6	Assertion (A): In the vacuum bell jar experiment, the sound of an electric bell becomes fainter as air is sucked out from the bell jar. Reason (R): Sound needs a medium to propagate and cannot travel through vacuum. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (A)	1



Q.NO.	QUESTIONS	MARKS
	► Why? A. The assertion is true because the source states that as air is sucked out from the bell jar, the sound becomes fainter. The reason is also true and correctly explains the assertion, as sound cannot propagate in vacuum and needs a medium to travel, which is why the sound fades as the medium (air) is removed. [Ch 10: Sound Waves: Characteristics and Applications]	
7	Imagine an object moving on a perfectly smooth horizontal floor where the force of friction between them is zero. What will happen to the object's velocity? (A) It will not decrease and the object will continue moving forever. (B) It will gradually decrease, but the object will never stop. (C) It will decrease at the same rate as on a rough surface. (D) It will instantly drop to zero. Answer: Correct Option: (A) ► Why? A. If all impediments like friction are removed, the object's velocity will not decrease and it will continue to move indefinitely. [Ch 6: How Forces Affect Motion]	1
8	Assertion (A): Cultured phloem cells of carrot in a liquid medium with nutrients show a 20% increase in fresh weight when supplied with air. Reason (R): The presence of light along with a solid medium and nutrients increases the biomass of the cultured plant cells. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (C) ► Why? C. The assertion is true because the experimental data shows that under air with a liquid nutrient medium, the fresh weight increased by 20%. The reason is false because light with a solid medium and nutrients resulted in reduced growth, not an increase in biomass. [Ch 3: Tissues in Action]	1
9	What is the SI unit of force and how is its symbol correctly written? (A) newton, written as N (B) Newton, written as n (C) newton, written as n (D) Newton, written as N Answer: Correct Option: (A) ► Why? A. The SI unit of force is newton (written with a small 'n') and its symbol is N (capitalised). [Ch 6: How Forces Affect Motion]	1



Q.NO.	QUESTIONS	MARKS
10	Sound is a longitudinal mechanical wave that: (A) Can travel through vacuum (B) Does not require a medium to propagate (C) Only travels through gases (D) Needs a medium to travel Answer: Correct Option: (D) ► Why? D. Sound is a longitudinal mechanical wave that needs a medium to travel; it cannot propagate through vacuum. [Ch 10: Sound Waves: Characteristics and Applications]	1
11	What is a likely consequence of errors in meiosis? (A) Genetic disorders and reduced fertility (B) Formation of tumours in body cells (C) Uncontrolled cell divisions (D) Abnormal number of chromosomes only in skin cells Answer: Correct Option: (A) ► Why? A. Errors in meiosis may result in genetic disorders associated with developmental problems, distinctive physical features, or reduced fertility. [Ch 2: Cell: The Building Block of Life]	1
12	Assertion (A): The path of a laser beam is not visible when passed through a true solution. Reason (R): The particles in a suspension scatter the beam of light, making its path visible. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (B) ► Why? B. The assertion is true because the particles in a solution are too small to scatter light. The reason is also true as suspension particles do scatter light to make the path visible, but it does not explain why the path is invisible in a true solution. [Ch 5: Exploring Mixtures and their Separation]	1



Q.NO.	QUESTIONS	MARKS
13	Assertion (A): The experiment conducted by J. Craig Venter and his team created a completely new cell from scratch. Reason (R): The synthetic DNA inserted into the cell controlled the structure and activities of the cell. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (D) ► Why? A is false but R is true. The Venter experiment did not create a completely new cell from scratch because only the DNA was synthetic while the rest of the cell was taken from an existing living cell. However, the experiment did show that DNA controls the structure and activities of a cell, making the reason true. [Ch 2: Cell: The Building Block of Life]	1
14	What is the limit of resolution of the human eye when viewing from a distance of about 25 cm? (A) 0.01 mm (B) 0.1 mm (C) 1.0 mm (D) 10 mm Answer: Correct Option: (B) ► Why? B. The limit of resolution of the human eye is 0.1 mm, meaning two points separated by about 0.1 mm can be seen as distinct when viewed from about 25 cm. [Ch 2: Cell: The Building Block of Life]	1
15	A solid that is made up of particles arranged in a regular geometric pattern is called: (A) An amorphous solid (B) A precipitate (C) A crystal (D) An alloy Answer: Correct Option: (C) ► Why? C. A crystal is a solid that is made up of particles arranged in a regular geometric pattern. [Ch 5: Exploring Mixtures and their Separation]	1



Q.NO.	QUESTIONS	MARKS
16	For an object moving in a straight line in the same direction, when is its acceleration considered constant? (A) When its velocity is zero (B) When it travels equal distances in equal intervals of time (C) When its displacement is zero (D) When the magnitude of its velocity increases or decreases by equal amounts in equal intervals of time Answer: Correct Option: (D) ► Why? D. Acceleration is constant if the magnitude of velocity increases or decreases by equal amounts in equal intervals of time. <i>[Ch 4: Describing Motion Around Us]</i>	1
17	If an object moving in a straight line travels equal distances in equal intervals of time (for all possible choices of time intervals), its motion is described as: (A) Uniform motion (B) Non-uniform motion (C) Accelerated motion (D) Oscillatory motion Answer: Correct Option: (A) ► Why? A. An object travelling equal distances in equal intervals of time is said to be in uniform motion in a straight line. <i>[Ch 4: Describing Motion Around Us]</i>	1
18	Why is blood fluid while bone is hard? (A) The matrix in blood is watery and soft, while in bones it is hard and rigid. (B) Blood lacks a matrix, while bone has a rigid matrix. (C) Blood contains only cells, while bones contain only fibres. (D) The matrix in blood is hard, while in bones it is jelly-like. Answer: Correct Option: (A) ► Why? A. The difference in consistency is due to the matrix, which is watery, soft and jelly-like in blood but hard, solid, and rigid in bones. <i>[Ch 3: Tissues in Action]</i>	1
19	Stainless steel is an alloy of iron that contains which of the following elements? (A) Only carbon and chromium (B) Copper (80%) and zinc (20%) (C) Copper (80%) and tin (20%) (D) Carbon (about 0.03–0.8%), chromium (16–18%), nickel (10.0–14.0%), and molybdenum (2.0–3.0%) Answer: Correct Option: (D)	1



Q.NO.	QUESTIONS	MARKS
	► Why? D. Stainless steel is a mixture of iron and other elements including carbon (about 0.03–0.8%), chromium (16–18%), nickel (10.0–14.0%), and molybdenum (2.0–3.0%), as specified in the source. [Ch 5: Exploring Mixtures and their Separation]	
20	Assertion (A): When a marble moving along the inner boundary of a ring is released by lifting the ring, it continues to move in a straight line. Reason (R): The velocity at a point on the circular path is along the tangent to the circle at that point, and the marble continues to move in the direction it has been moving at the instant of release. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (A) ► Why? A. The assertion is true because upon release, the marble loses the centripetal force and travels in a straight line. The reason correctly explains this by stating that the instantaneous velocity is directed along the tangent, so the marble continues in that straight-line tangent direction when released. [Ch 4: Describing Motion Around Us]	1
Section B Attempt 5 out of 7.		
21	State the significance of mitosis in multicellular organisms. Answer: Mitosis is important for normal growth, repair of damaged tissues, maintenance, and asexual reproduction. It produces two genetically identical daughter cells from one parent cell, ensuring genetic information is largely maintained across body cells. ► Why? The student must recall the biological importance of mitosis for growth and repair, as well as its outcome regarding the genetic identity of daughter cells. [Ch 2: Cell: The Building Block of Life]	2
	OR Differentiate between sublimation and deposition based on the state changes involved. Answer: Sublimation is the process where a solid changes directly into the vapour state without passing through the liquid state upon heating. Deposition is the reverse process where vapours condense back into a solid without becoming a liquid on cooling. ► Why? The source defines sublimation as a solid-to-vapour change without becoming liquid, and deposition as vapour condensing back to a solid without becoming liquid. [Ch 5: Exploring Mixtures and their Separation]	



Q.NO.	QUESTIONS	MARKS
22	A naval sonar signal sent into seawater returns after 0.90 s. If the speed of sound in seawater is 1530 m s^{-1}, how far is the object? Answer: Time taken for the signal to reach the object = $0.90/2 = 0.45 \text{ s}$. Distance = $1530 \text{ m s}^{-1} \times 0.45 \text{ s} = 688.5 \text{ m}$. The object is 688.5 m away from the sonar. ► Why? The total return time includes travel to the object and back, so the one-way time is half. Multiplying one-way time by speed yields the distance. [Ch 10: Sound Waves: Characteristics and Applications]	2
	OR A fielder catching a fast-moving cricket ball gradually pulls their hands backwards. Explain how this reduces the chance of injury. Answer: Pulling hands backwards increases the time duration during which the ball's high velocity reduces to zero. This reduces the magnitude of acceleration of the ball, thus requiring a smaller force to stop it, which minimises injury to the fielder. ► Why? Since $F = ma$, reducing acceleration (by increasing the time over which velocity changes) reduces the force the fielder must apply, thereby lowering injury risk. [Ch 6: How Forces Affect Motion]	
23	What is the thickness of the cell membrane and what is it made of? Answer: The cell membrane is extremely thin, about 7 to 10 nanometres (nm) thick. It is made up of lipids (fats) and proteins, arranged as a lipid bilayer with proteins embedded in it, as explained by the fluid-mosaic model. ► Why? The source provides the specific thickness (7-10 nm) and composition (lipids and proteins in a bilayer) of the cell membrane. [Ch 2: Cell: The Building Block of Life]	2
24	What does a straight line parallel to the time axis on a position-time graph indicate about the nature of motion of the object? Answer: A straight line parallel to the time axis on a position-time graph indicates that the object is stationary or at rest. This means the position of the object is not changing with time. ► Why? This requires interpreting a specific shape on a position-time graph, linking the horizontal line to zero velocity and a stationary object. [Ch 4: Describing Motion Around Us]	2
25	What is the primary function of tendons in the body, and how do they interact with the muscular and skeletal systems? Answer: Tendons are made of tough connective tissue that transmits force from muscle to bone. They interact with the musculoskeletal system by enabling the coordination of muscles and bones, which produces movement under the control of the nervous system.	2



Q.NO.	QUESTIONS	MARKS
	► Why? The text clarifies that tendons transmit force from muscles to bones, refuting the false assertion that they connect bone to bone, and highlights their role in the musculoskeletal system. [Ch 3: Tissues in Action]	
Section C Attempt 5 out of 7.		
26	How can a spring balance be used to compare the force of friction on a wooden block across different surfaces? What does a smaller reading indicate? Answer: A spring balance is attached to a wooden block and pulled horizontally across different surfaces. The reading on the spring balance is noted at the exact moment the block just starts moving. This reading gives an approximate measure of the force of friction. By repeating this process on various surfaces, the readings can be compared. A smaller reading on the spring balance indicates a smaller force of friction acting between the block and the surface. This means the velocity of the object would decrease more slowly, allowing it to travel a larger distance before coming to rest. ► Why? The answer explains the practical method of measuring friction using a spring balance and interprets the meaning of the reading based on the activity described in the text. ► Step-by-step: 1. Step 1: Attach the wooden block to the spring balance and pull it horizontally. 2. Step 2: Note the reading when the block just starts moving, which indicates the force of friction. 3. Step 3: Compare readings across surfaces; a smaller reading means a smaller force of friction. [Ch 6: How Forces Affect Motion]	3
	OR How does the epidermis protect plants from various environmental factors, and what role does the cuticle play in plants living in dry habitats? Answer: The epidermis forms the outermost layer of the plant body, consisting of a tightly packed, single layer of flat and rectangular cells. It protects all parts of the plant from mechanical injury, water loss, harmful microorganisms, and extreme environmental conditions. In plants living in very dry habitats, the epidermis is covered by a thick waxy layer of cutin called the cuticle. This thick cuticle significantly reduces water loss during transpiration through the stomata. Additionally, the cuticle provides protection against mechanical injury and invasion by parasites, ensuring the plant's survival in harsh, arid environments.	



Q.NO.	QUESTIONS	MARKS
	► Why? The epidermis acts as a protective barrier, while the cuticle, a waxy layer, minimizes water loss and protects against injury and parasites, which is crucial for survival in dry habitats. ► Step-by-step: - Step 1: Describe the epidermis as a tightly packed outermost layer providing general protection. - Step 2: Identify the cuticle as a waxy layer of cutin on the epidermis. - Step 3: Explain that in dry habitats, a thick cuticle reduces water loss by transpiration and protects against injury and parasites. [Ch 3: Tissues in Action]	
27	What is pitch? How does the pitch of a sound relate to its frequency, and what changes occur in the vocal cords of boys during adolescence that affect their voice pitch? Answer: Pitch is how humans perceive the frequency of a sound. Sounds that are shrill, like a whistle, have a high pitch, while deep sounds, like thunder, have a low pitch. In general, high pitch sounds have a higher frequency and low pitch sounds have a lower frequency. During adolescence, the vocal cords of boys lengthen and thicken. Because of this physical change, their vocal cords vibrate less frequently. This decrease in vibration frequency causes their voice to deepen, resulting in a lower pitch compared to children or some females. ► Why? Pitch is the subjective perception of frequency. Thicker and longer vocal cords vibrate at lower frequencies, producing a deeper pitch. ► Step-by-step: - Step 1: Define pitch as the human perception of frequency (high pitch = high frequency, low pitch = low frequency). - Step 2: Explain the physical change in boys during adolescence (vocal cords lengthen and thicken). - Step 3: Connect the physical change to frequency (vibrate less frequently) and the resulting pitch (voice deepens). [Ch 10: Sound Waves: Characteristics and Applications]	3
	OR Sewage water treatment involves several steps to clean the water before it is released into the environment. Explain the role of coagulation in this process and state one way the clean water obtained can be reused. Answer: Coagulation is a crucial step in sewage treatment that involves the addition of a substance called a coagulant. The coagulant causes the smaller particles in the heterogeneous mixture of solids and liquids to clump together and settle down. This makes it easier to remove the solid waste from the water. Once the sewage water is treated and cleaned through steps like sedimentation, coagulation, and filtration, it can be safely reused for non-drinking purposes. For example, the clean water obtained can be reused for flushing toilets or watering plants.	



Q.NO.	QUESTIONS	MARKS
	► Why? Coagulation uses a coagulant to make small particles clump together and settle, facilitating their removal from sewage water. ► Step-by-step: 1. Step 1: Define coagulation as the addition of a coagulant to a heterogeneous mixture. 2. Step 2: Explain that the coagulant makes smaller particles clump together and settle down. 3. Step 3: State the reuse applications for the clean water obtained, such as flushing toilets or watering plants. [Ch 5: Exploring Mixtures and their Separation]	
28	A car starts from rest and its velocity reaches 24 m s^{-1} in 6 s. Find the average acceleration and the distance travelled in these 6 s. Answer: Given: Initial velocity, $u = 0 \text{ m s}^{-1}$; Final velocity, $v = 24 \text{ m s}^{-1}$; Time, $t = 6 \text{ s}$ Formula: Average acceleration, $a = (v - u)/t$ Distance, $s = ut + 1/2at^2$ Calculation: $a = (24 - 0)/6 = 4 \text{ m s}^{-2}$ $s = (0 \times 6) + 1/2(4)(6)^2 = 0 + 72 = 72 \text{ m}$ ∴ The average acceleration is 4 m s^{-2} and the distance travelled is 72 m. These values are calculated using the definition of average acceleration and the kinematic equations for motion in a straight line with constant acceleration. ► Why? The average acceleration is found using the change in velocity over time, and the distance is calculated using the kinematic equation for constant acceleration. ► Step-by-step: 1. Step 1: Use the definition of average acceleration: $a = (v - u) / t$, substituting $v = 24$, $u = 0$, $t = 6$. 2. Step 2: Calculate $a = (24 - 0) / 6 = 4 \text{ m s}^{-2}$. 3. Step 3: Use the kinematic equation $s = ut + 1/2at^2$ to find distance, substituting $u = 0$, $a = 4$, $t = 6$. 4. Step 4: Calculate $s = 0 + 1/2(4)(36) = 72 \text{ m}$. [Ch 4: Describing Motion Around Us]	3
29	Differentiate between the following pairs of terms based on the clues given in parentheses: (i) Cell membrane and cell wall (permeability) (ii) RER and SER (structure) (iii) Chloroplasts and chromoplasts (pigments)	3



Q.NO.	QUESTIONS	MARKS
	Answer: (i) (i) The cell membrane is selectively permeable, which means it allows only specific molecules to pass through it. In contrast, the cell wall is fully permeable, which means it allows all substances to pass through it freely. (ii) (ii) The RER (Rough Endoplasmic Reticulum) has ribosomes attached to its surface, which gives it a rough structure. In contrast, the SER (Smooth Endoplasmic Reticulum) lacks ribosomes on its surface, which gives it a smooth structure. (iii) (iii) Chloroplasts contain the green pigment chlorophyll, which is essential for photosynthesis. Chromoplasts contain coloured pigments (other than green), such as yellow, orange, or red pigments. ► Why? The question requires differentiating between three pairs of cellular components based on specific clues. For permeability, the cell membrane acts as a selective barrier, whereas the cell wall allows free passage. Structurally, the presence or absence of ribosomes distinguishes RER from SER. Based on pigments, chloroplasts are identified by chlorophyll, while chromoplasts contain other coloured pigments. [Ch 2: Cell: The Building Block of Life]	
30	What is Programmed Cell Death (PCD)? Explain its role in human embryonic development. Answer: Programmed Cell Death (PCD) is a genetically regulated and organised process of selective cell destruction. It is a natural way cells die to maintain a balance in the body. PCD is essential for normal development, cellular quality control, and immune function. During embryonic development, PCD plays a crucial role in shaping the body. For instance, it helps form fingers by eliminating the cells between the digits. Without this programmed elimination of cells, humans would have webbed hands. Therefore, PCD is vital for proper structural formation and removing cells that are no longer needed. ► Why? The answer defines PCD as a genetically regulated process of selective cell destruction and explains its specific role in eliminating cells between digits to prevent webbed hands. ► Step-by-step: 1. Step 1: Define PCD as a genetically regulated process of selective cell destruction. 2. Step 2: Explain its general role in normal development, quality control, and immune function. 3. Step 3: Describe its specific role in forming fingers by eliminating cells between digits, preventing webbed hands. [Ch 2: Cell: The Building Block of Life]	3
Section D Attempt 4 out of 6.		
31	Describe the step-by-step procedure for plotting a position-time graph from given data. In your explanation, include how to draw the axes, choose suitable scales, plot the data points, and connect them to form the graph.	5



Q.NO.	QUESTIONS	MARKS
	Answer: To plot a position-time graph, the first step is to take a sheet of graph paper, which is pre-divided into small squares, making it easier to plot data accurately. On this paper, draw two lines perpendicular to each other. Their point of intersection is known as the origin O. Mark the horizontal line as OX (x-axis) and the vertical line as OY (y-axis). Next, decide which quantity to show along each axis. For motion data, time is shown along the x-axis and position along the y-axis. Then, determine a suitable scale for each quantity so that the data is represented effectively, conveniently, and utilises the available space. For example, the scale could be x-axis: 5 divisions = 1 s and y-axis: 5 divisions = 20 m. Use the chosen scale to mark values for time along the x-axis and for position along the y-axis from the origin. Now, begin plotting points for each set of time and position values. For example, at time 0 s, if position is 0 m, the point is the origin itself. At 1 s and 20 m, draw a line parallel to the y-axis at 1 s on the x-axis and a line parallel to the x-axis from 20 m on the y-axis; their intersection is the required point. Similarly, plot all points corresponding to positions at different instants of time. Once all points are plotted, connect them to create the position-time graph. If the object moves with constant speed, the result is a straight line; if it speeds up, the points form a curve. ► Why? Plotting a graph involves drawing perpendicular axes, choosing scales, marking values, plotting each data point by finding intersections of parallel lines from both axes, and then connecting the points. ► Step-by-step: 1. Step 1: Describe setting up graph paper and drawing perpendicular axes (x-axis and y-axis) with origin O. 2. Step 2: Explain assigning time to x-axis and position to y-axis. 3. Step 3: Explain choosing suitable scales for both axes with an example. 4. Step 4: Describe how to plot each point using lines parallel to the axes and finding their intersection. 5. Step 5: Explain connecting all plotted points to complete the graph and note the shape obtained. [Ch 4: Describing Motion Around Us]	
	OR What are emulsions? Differentiate between oil-in-water and water-in-oil emulsions with examples. Explain the role of emulsifying agents and describe how you can prepare an emulsion using cooking oil and water.	



Q.NO.	QUESTIONS	MARKS
	Answer: Emulsions are a special type of colloid where both the dispersed phase and the dispersion medium are liquids. In an emulsion, tiny droplets of one liquid are dispersed throughout another liquid, forming a colloidal mixture that would normally separate without stabilization. Emulsions are classified into two main types: oil-in-water emulsions and water-in-oil emulsions. In oil-in-water emulsions, oil acts as the dispersed phase and water acts as the dispersion medium; common examples include milk and vanishing creams. In water-in-oil emulsions, water acts as the dispersed phase and oil acts as the dispersion medium; examples include butter, body lotions, and cold creams. Emulsifying agents are substances that stabilize emulsions by preventing the dispersed liquid droplets from coming together and separating. Without these emulsifying agents, the two immiscible liquids would separate into distinct layers. For example, proteins in milk and butter act as natural emulsifying agents, keeping the mixture stable over time. You can prepare an emulsion at home by thoroughly mixing and shaking a few drops of cooking oil with water containing a few drops of soap solution. The soap acts as an emulsifying agent, allowing the oil to disperse in the water and forming a stable emulsion. Some medicines are also prepared as emulsions to disperse them in water, making them easily palatable and reducing the greasy feeling of the liquid medicine. ► Why? Emulsions are liquid-in-liquid colloids stabilized by emulsifying agents, classified based on which liquid is the dispersed phase versus the dispersion medium. ► Step-by-step: 1. Step 1: Define emulsions as colloids where both the dispersed phase and dispersion medium are liquids. 2. Step 2: Differentiate oil-in-water (milk, vanishing cream) and water-in-oil (butter, cold cream) emulsions. 3. Step 3: Explain the role of emulsifying agents (like proteins in milk) in stabilizing emulsions. 4. Step 4: Describe the preparation of an emulsion using cooking oil, water, and soap solution. 5. Step 5: Mention the application of emulsions in medicines for palatability. [Ch 5: Exploring Mixtures and their Separation]	
32	Describe the vacuum bell jar experiment to demonstrate that sound needs a medium to propagate. What observations are made when air is removed and then let back in? Also, explain how this principle applies to astronauts on spacewalks.	5



Q.NO.	QUESTIONS	MARKS
	Answer: Sound requires a medium to propagate from its source to the listener. The material through which sound propagates is called a medium, and it can be a solid, a liquid, or a gas. A space where there is no medium or matter is referred to as a vacuum. The vacuum bell jar experiment demonstrates that sound needs a medium. An electric bell is kept inside a bell jar and switched on, and the loudness of the sound is noted. As air is sucked out from the bell jar using a vacuum pump, the sound becomes fainter. Once a near vacuum is reached inside the jar, almost no sound can be heard, even though the bell can still be seen ringing. When air is let back into the jar, the sound can be heard again and gradually becomes as loud as before. This experiment leads to the conclusion that sound cannot propagate in vacuum. Sound fundamentally needs a medium to travel, and without matter, it cannot move from one place to another. This principle explains why astronauts on spacewalks cannot directly hear each other speak or hear sounds like two metal objects clanking together. Because outer space is a near vacuum, there is no medium for sound to travel through. Therefore, astronauts must communicate through special devices fitted into their spacesuits. ► Why? The vacuum bell jar experiment proves that sound requires a material medium to travel, as removing the air stops the sound despite the bell still ringing, which directly explains the need for communication devices in the vacuum of outer space. ► Step-by-step: 1. Step 1: Define what a medium and a vacuum are in the context of sound propagation. 2. Step 2: Describe the setup of the vacuum bell jar experiment with the electric bell. 3. Step 3: Explain the observations when air is removed using a vacuum pump and when air is let back in. 4. Step 4: State the conclusion drawn from the experiment about sound needing a medium and not propagating in a vacuum. 5. Step 5: Apply the conclusion to explain why astronauts need special devices in the near vacuum of outer space. [Ch 10: Sound Waves: Characteristics and Applications]	
	OR Explain the process of osmosis and how it differs from diffusion. Describe the behaviour of cells when placed in isotonic, hypotonic, and hypertonic solutions, using the example of potato pieces in plain water and salt solution.	



Q.NO.	QUESTIONS	MARKS
	Answer: Osmosis is the diffusion of water across a selectively permeable membrane. Water moves from an area with more water and less solute to an area with less water and more solute until the concentrations in the two areas become equal. Diffusion, in contrast, is the net movement of particles from a higher to a lower concentration and occurs even without a membrane. When a potato piece is placed in plain water (Beaker A), water moves into the potato cells by osmosis because the extracellular medium has less solute than the intracellular medium. This is a hypotonic solution. The potato piece swells and its weight increases. When a potato piece is placed in a 20 per cent salt solution (Beaker B), water moves out of the potato cells because the extracellular medium has more solute than the intracellular medium. This is a hypertonic solution. The potato piece shrinks and its weight decreases. In an isotonic solution, the solute concentration of the extracellular medium equals the solute concentration of the intracellular medium, so there is no net movement of water and the cell maintains its normal state. The key difference between osmosis and diffusion is that osmosis requires a selectively permeable membrane and involves only water movement, while diffusion does not require a membrane and involves the movement of any particles from higher to lower concentration. ► Why? Osmosis is water movement across a selectively permeable membrane driven by concentration differences; cells swell in hypotonic solutions and shrink in hypertonic solutions. ► Step-by-step: 1. Step 1: Define osmosis and diffusion, and state the key difference between them. 2. Step 2: Explain the potato experiment — swelling in plain water (hypotonic) and shrinking in salt solution (hypertonic). 3. Step 3: Define isotonic, hypotonic, and hypertonic solutions with their effects on cells. 4. Step 4: Compare osmosis and diffusion clearly — membrane requirement and type of particles moving. 5. Step 5: Conclude summarising how solution type determines the direction of water movement. [Ch 2: Cell: The Building Block of Life]	
33	Describe how the force of friction affects the motion of an object on different surfaces. Explain the investigation using a stack of coins and a rubber band on surfaces such as a wooden table top, a laminated table top, and a polished marble floor, and state the conclusions drawn about the relationship between friction and the distance travelled.	5



Q.NO.	QUESTIONS	MARKS
	Answer: Force of friction acts on an object in the direction opposite to its motion. When an applied force on a moving object is removed, the force of friction gradually decreases the velocity of the object and finally brings it to rest. In the investigation, a stack of four coins is secured with adhesive tape and placed on a surface. A rubber band stretched between the forefinger and thumb at marked points A and B is pulled back to a mark C, and the stack is released. The rubber band applies a forward force greater than friction, creating a net force that accelerates the coins forward. Once the coins lose contact with the rubber band, only friction acts, gradually slowing them to rest. The distance travelled from point C is measured. This is repeated on three surfaces: wooden table top, laminated table top, and polished marble or tiled floor. The stack travels the shortest distance on the wooden table top and the largest distance on the polished marble floor. Its velocity decreases more slowly on the smoother surface. A spring balance attached to a wooden block confirms that the reading — which measures the force of friction — is smallest on the surface where the coins travelled the largest distance and largest on the surface where they travelled the shortest distance. The conclusion is that when the force of friction is smaller, the velocity of the object decreases more slowly and it travels a larger distance before coming to rest. The force of friction depends upon the nature of the surfaces in contact. ► Why? Friction opposes motion and varies with surface type; smoother surfaces produce less friction, allowing objects to travel farther before stopping. ► Step-by-step: 1. Step 1: Explain that friction opposes motion and brings moving objects to rest. 2. Step 2: Describe the setup — coins released from a stretched rubber band on different surfaces, measuring distance travelled. 3. Step 3: State observations — shortest distance on wooden table, largest on polished marble, with velocity decreasing more slowly on smoother surfaces. 4. Step 4: Explain the spring balance measurement confirming different friction on different surfaces. 5. Step 5: Conclude that smaller friction means slower velocity decrease and larger distance before rest. [Ch 6: How Forces Affect Motion]	
34	Define totipotency. Describe how F.C. Steward demonstrated this property using carrot phloem cells, and explain the processes of dedifferentiation and redifferentiation that occur during the regeneration of a complete plant.	5



Q.NO.	QUESTIONS	MARKS
	Answer: Totipotency is the ability of some mature plant cells to undifferentiate, divide, and redifferentiate to develop into a new plant under specific conditions. This is similar to the ability of a zygote to divide and differentiate into an entire organism. In 1958, F.C. Steward and his team demonstrated totipotency using phloem cells of carrot. They grew these phloem cells in a nutrient medium containing simple sugars and hormones under appropriate conditions. The phloem cells divided to form a mass of cells, which gradually divided and differentiated into a complete plant. The process involves two key stages. First, the mature phloem cells dedifferentiate — they regain the ability to divide — forming an undifferentiated mass of unspecialised cells. Second, when these unspecialised cells are grown in appropriate conditions supplemented with nutrients and growth chemicals, they further divide and redifferentiate to form roots, shoot, and eventually the complete plant. The experimental results also showed that growth conditions matter. Cells grown in a liquid medium with nutrients and air showed a 20% increase in fresh weight, while cells under other conditions showed reduced growth. This confirms that totipotency requires specific conditions of nutrients, air, and growth chemicals to be expressed fully. ► Why? Totipotency is demonstrated by Steward's experiment where carrot phloem cells dedifferentiated into unspecialised cells and then redifferentiated into a complete plant, proving mature plant cells can regenerate under appropriate conditions. ► Step-by-step: 1. Step 1: Define totipotency as the ability of mature plant cells to undifferentiate, divide and redifferentiate into a new plant under specific conditions. 2. Step 2: Describe Steward's experiment — phloem cells of carrot grown in nutrient medium with simple sugars and hormones formed a complete plant. 3. Step 3: Explain dedifferentiation — mature cells regain the ability to divide, forming an undifferentiated mass of unspecialised cells. 4. Step 4: Explain redifferentiation — unspecialised cells, under appropriate conditions with nutrients and growth chemicals, divide and differentiate into roots, shoot, and complete plant. 5. Step 5: Conclude by mentioning that specific conditions (liquid medium, air, nutrients) are essential, as shown by the 20% increase in fresh weight versus reduced growth under other conditions. [Ch 3: Tissues in Action]	
Section E Attempt 3 out of 4.		



Q.NO.	QUESTIONS	MARKS
35	Case Study: Plant Protective and Supporting Tissues A botanist comparing plants from different habitats observes that desert plants have a thick waxy coating on their surface, while aquatic plants float easily on water. She notices that young plant stems bend without breaking, but the hard shells of walnuts are extremely tough and cannot bend. She examines the transverse section of a sunflower stem under a microscope and identifies several types of permanent tissues. She finds that permanent tissues can be simple (composed of only one type of cell) or complex (composed of more than one type of cell), and each tissue is specialised to perform a specific function such as support, transport, or storage. (i) What is the waxy layer of cutin on the epidermis called? (ii) Which type of simple permanent tissue allows plant stems and tendrils to bend without breaking? Name the chemical responsible for this flexibility. (iii) A walnut shell is hard and strong. Identify the tissue, the chemical deposited in its cell walls, and state whether its cells are living or dead. (iv) How does the specialised parenchyma in aquatic plants help them survive in their habitat? Answer: (i) Cuticle (ii) Collenchyma; pectin (a chemical that gives flexibility like rubber) is deposited at the unevenly thickened corners. (iii) Sclerenchyma; lignin is deposited in its cell walls; most of these cells are dead. (iv) Specialised parenchyma forms air spaces which help aquatic plants float. ► Why? i) Cuticle ii) Collenchyma; pectin (a chemical that gives flexibility like rubber) is deposited at the unevenly thickened corners. iii) Sclerenchyma; lignin is deposited in its cell walls; most of these cells are dead. iv) Specialised parenchyma forms air spaces which help aquatic plants float. [Ch 3: Tissues in Action]	5



Q.NO.	QUESTIONS	MARKS
	OR Case Study: Concentration of Solutions in Everyday Products A student examines three products. A packaged food item contains 8 g of salt in 200 g of the product. A glucose solution bottle from a pharmacy is labelled 5% m/v, meaning 5 g of glucose is present in 100 mL of the solution. A milk powder packet lists its protein content as 15% m/m. The student notes that mass by mass percentage is commonly used for heterogeneous mixtures like milk powder, while mass by volume percentage is used in medicines and laboratories where measuring the volume of a liquid is easier than weighing it. (i) What does 5% m/v on the glucose solution bottle indicate? (ii) Why is mass by volume percentage used for medicines like glucose solution instead of mass by mass percentage? (iii) Calculate the mass by mass percentage of salt in the packaged food item if it contains 8 g of salt in 200 g of the product. (iv) Milk powder is labelled using mass by mass percentage for its composition. What type of mixture is milk powder, and why is this method of expressing concentration appropriate for it? Answer: (i) 5% m/v means that 5 grams of glucose is present in 100 millilitres of the solution. (ii) Mass by volume percentage is used because measuring the volume of a liquid is easier than weighing it in medicines and laboratories. (iii) Mass of solute (salt) = 8 g, Mass of solution = 200 g. Mass by mass percentage = (Mass of solute / Mass of solution) × 100 = (8/200) × 100 = 4% m/m. (iv) Milk powder is a heterogeneous mixture. Mass by mass percentage is appropriate because this method is also used for heterogeneous mixtures such as milk powder and spice mixtures. ► Why? i) 5% m/v means that 5 grams of glucose is present in 100 millilitres of the solution. ii) Mass by volume percentage is used because measuring the volume of a liquid is easier than weighing it in medicines and laboratories. iii) Mass of solute (salt) = 8 g, Mass of solution = 200 g. Mass by mass percentage = (Mass of solute / Mass of solution) × 100 = (8/200) × 100 = 4% m/m. iv) Milk powder is a heterogeneous mixture. Mass by mass percentage is appropriate because this method is also used for heterogeneous mixtures such as milk powder and spice mixtures. [Ch 5: Exploring Mixtures and their Separation]	



Q.NO.	QUESTIONS	MARKS
36	Case Study: Average Acceleration When a bus moves on a long straight highway, its velocity can change. The average acceleration of the bus over a time interval is the change in its velocity divided by the time interval. For instance, if a driver presses the accelerator, the velocity increases, and the average acceleration is in the direction of velocity. Conversely, if the driver presses the brake, the velocity decreases, and the average acceleration is opposite to the direction of velocity. Interestingly, a bus moving at a constant velocity, even if it is very fast, has zero acceleration because acceleration depends on how quickly velocity changes. (i) What is the SI unit of average acceleration? (ii) A bus moves on a straight highway at a constant velocity. What is its average acceleration? (iii) A bus increases its velocity from 10 m s^{-1} to 15 m s^{-1} in a time interval of 10 s. Calculate the average acceleration. (iv) When the magnitude of velocity of a bus is decreasing, in which direction does the average acceleration act relative to the velocity? Answer: (i) m s^{-2} or m/s^2. (ii) Zero acceleration. (iii) Average acceleration = (Final velocity - Initial velocity) / Time interval = $(15 - 10) / 10 = 0.5 \text{ m s}^{-2}$. (iv) Opposite to the direction of velocity. ► Why? i) m s^{-2} or m/s^2. ii) Zero acceleration. iii) Average acceleration = (Final velocity - Initial velocity) / Time interval = $(15 - 10) / 10 = 0.5 \text{ m s}^{-2}$. iv) Opposite to the direction of velocity. [Ch 4: Describing Motion Around Us]	5
37	Case Study: Energy and Food Synthesis in Cells Plants and animals have specialised organelles to meet their energy and nutritional needs. In animal cells, the 'powerhouses' generate energy through cellular respiration, breaking down glucose to produce ATP. These organelles have an outer smooth, porous membrane and an inner membrane folded into cristae. Meanwhile, plant cells contain double-membrane-bound organelles that synthesise food using sunlight. Inside these organelles, a semi-fluid substance called the stroma contains disc-shaped structures that hold the green pigment chlorophyll. Interestingly, both types of organelles possess their own DNA and ribosomes, allowing them to synthesise some of their own proteins. These features suggest they share an evolutionary history with single-celled organisms. (i) What is the function of the cristae present in the inner membrane of mitochondria? (ii) Why is ATP referred to as the 'energy currency' of the cell? (iii) Describe the role of the stroma and the disc-shaped membrane structures inside the chloroplast during photosynthesis. (iv) Based on their characteristics, what can be inferred about the evolutionary history of mitochondria and plastids?	5



Q.NO.	QUESTIONS	MARKS
	Answer: (i) Cristae increase the surface area for chemical reactions and facilitate energy production. (ii) The energy released during cellular respiration is stored in the form of ATP, which is used for most cellular activities. (iii) The disc-shaped membrane structures contain chlorophyll, which absorbs sunlight (light energy). The sugars synthesised during photosynthesis are stored in the stroma along with starch granules. (iv) Both mitochondria and plastids share an evolutionary history with single-celled organisms (bacteria) because they have their own DNA and ribosomes. ► Why? i) Cristae increase the surface area for chemical reactions and facilitate energy production. ii) The energy released during cellular respiration is stored in the form of ATP, which is used for most cellular activities. iii) The disc-shaped membrane structures contain chlorophyll, which absorbs sunlight (light energy). The sugars synthesised during photosynthesis are stored in the stroma along with starch granules. iv) Both mitochondria and plastids share an evolutionary history with single-celled organisms (bacteria) because they have their own DNA and ribosomes. [Ch 2: Cell: The Building Block of Life]	

